

La norma viene de un producto interno si y solo si satisface la identidad del paralelogramo

Teorema 1. Sea X un espacio vectorial con norma $\|\cdot\|$, entonces

$\exists \langle \cdot, \cdot \rangle$ producto interior que induce $\|\cdot\| \iff \|\cdot\|$ satisface la identidad del paralelogramo.

Demostración: Veamos ambas implicaciones.

$\Rightarrow$ Se sigue de la **identidad-paralelogramo**/Proposición 1.

$\Leftarrow$ Definimos $\forall x, y \in X : \langle x, y \rangle := \frac{1}{4} (\|x + y\|^2 - \|x - y\|^2 + i \|x + iy\|^2 - i \|x - iy\|^2)$ para que se satisfaga la identidad de polarización. Veamos que es un producto interno:

(i) Sean $x, y, z \in X$

$$\begin{aligned} \langle x, z \rangle + \langle y, z \rangle &= \frac{1}{4} (\|x + z\|^2 - \|x - z\|^2 + i \|x + iz\|^2 - i \|x - iz\|^2) \\ &\quad + \frac{1}{4} (\|y + z\|^2 - \|y - z\|^2 + i \|y + iz\|^2 - i \|y - iz\|^2) \\ &= \frac{1}{4} \left(\|(x + y) + z\|^2 - \|(x + y) - z\|^2 \right. \\ &\quad \left. + i \|(x + y) + iz\|^2 - i \|(x + y) - iz\|^2 \right) = \langle x + y, z \rangle. \end{aligned}$$

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Referenciado en

- cor-norma-p-no-prod-interno
- teo-l2-unico-esp-hilbert

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